

2024



Crossuite Saves 30% on Costs and Achieves 99.9% Uptime Using Amazon EKS

Learn how Crossuite modernized its next-gen eHealth application using Amazon EKS, improving scalability to support over 13 million patients

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30%

reduction in overall costs

99%

uptime achieved post migration

22%

YoY revenue growth achieved

18% user growth

YoY, serving 13 million users seamlessly



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Overview

Over 10,000 health professionals throughout Europe rely on the cloud-based practice-management platform from Crossuite to support their day-to-day tasks and treat their patients. For Crossuite, this means that maintaining



high availability is critical. The company wanted its solution to scale to meet growing demand, so it chose to modernize using Amazon Web Services (AWS). Thus, it improved performance and availability while reducing costs and operational overhead, resulting in an 18 percent increase in user adoption and 22 percent year-over-year (YoY) revenue growth.



Opportunity | Migrating to Amazon EKS to Meet Growing Demand

Founded in 2007, Belgian health-tech scale-up Crossuite is a pioneer in cloud-based practice-management solutions. Its all-in-one solution offers electronic medical records, financial operations, patient portals, automated communications, scheduling, and a mobile app. The solution is designed to streamline workflows and enhance patient interactions and is widely adopted in the healthcare industry. As a result of its significant growth, Crossuite now serves over 10,000 health practitioners and supports more than 13 million patients across over 23 countries.

Originally built on on-premises infrastructure, Crossuite managed its Kubernetes clusters manually. This process was time-consuming and diverted developers from more critical tasks, such as enhancing the user experience, improving system security, and developing new features. The solution, which is designed for healthcare professionals, saw traffic spikes during business days and quieter periods during evenings and weekends, complicating cluster management. The need to dynamically adjust to varying loads for optimal performance and resource efficiency necessitated a robust, flexible infrastructure. “We don’t want to run at full capacity when we don’t have to,” says Tomas Madliak, cloud technology consultant at Crossuite. Crossuite faced challenges with its on-premises infrastructure, including manual Kubernetes cluster management that consumed developer time and hindered focus on innovation.

In 2021, Crossuite recognized that modernization was the key to future growth. “We wanted to become more agile and resilient and to future-proof our critical applications by being cloud based,” says Madliak. Meeting scalability needs for the growing demand of the solution was crucial, with a major focus on designing an architecture to accommodate future user-base growth. Crossuite worked alongside AWS to set up a proof of concept, starting with lifting and shifting its legacy infrastructure to [Amazon Elastic Kubernetes Service](#) (Amazon EKS), a managed Kubernetes service. AWS recommended testing automatic scaling using [Karpenter](#), an open-source Kubernetes cluster automatic scaler, alongside Amazon EKS, to establish a dynamic infrastructure capable of scaling on demand

Solution | Achieving 99.9% Uptime While Scaling to Millions of Application Users

As Crossuite onboarded more clients, it aimed to scale effectively, reduce compute costs, and free up its development teams. The goal was to enhance its infrastructure without impacting the application, focusing instead on monitoring, testing, and refining the setup during migration. Crossuite used the AWS best-practices framework for migrating to Amazon EKS, covering external databases, APIs, third-party services, and necessary network adjustments. It was crucial to adapt security policies, identity and access management roles, and compliance checks to AWS using the robust security controls of Amazon EKS.

Crossuite designed its architecture with scalability and high availability in mind. Using Terraform, the team provisioned the Amazon EKS cluster, while GitLab's continuous integration and deployment pipelines streamlined application deployment onto the Amazon EKS environment. They enabled Horizontal Pod Autoscaling (HPA) for customer-facing services to ensure dynamic scaling. Crossuite configured the ingress layer with an AWS Application Load Balancer, which provided flexible routing and load distribution. For the backend, the team used [AWS Relational Database Service](#) (RDS) PostgreSQL, taking advantage of its scalability and reliability.

The company used observability and monitoring tools such as [Amazon CloudWatch](#)—a service that monitors applications, responds to performance changes, optimizes resource use, and provides insights into operational health. Thus, Crossuite was able to track application performance and health during and after the migration. With this level of observability, Crossuite confidently migrated massive amounts of data quickly and switched production traffic to Amazon EKS. “We wanted to be validated by AWS solution architects so that we didn’t miss anything,” says Lukas Pollak, DevOps and server reliability engineer at Crossuite. “We got really useful advice from AWS.”

Crossuite seamlessly switched its new Amazon EKS clusters to production in May 2024. This migration removed many manual tasks for the development team, such as spinning up new nodes, manually configuring the scaling policies and mechanisms, managing complex networking and redundancy planning across multiple servers, and redistributing pods across them. The company has multiple pods running on Amazon EKS, and it adopted Karpenter to improve application availability and cluster efficiency by rapidly launching rightsized compute resources in response to the changing application load. Previously, it took the Crossuite team around 15 minutes to spin up a new node. Now, using Karpenter, this can be done automatically in less than 2 minutes—that is, about an 86 percent reduction in time. “It’s seamless for us,” says Madliak. “Plus, the huge flexibility in this setup prepares us for the future.”

Crossuite also established a continuous integration and continuous deployment pipeline on AWS to roll out cluster updates with one click. “Our team can now focus more on development and innovations, not maintenance,” says Pollak. “The overall benefit is a more robust, efficient, and cost-effective solution.”

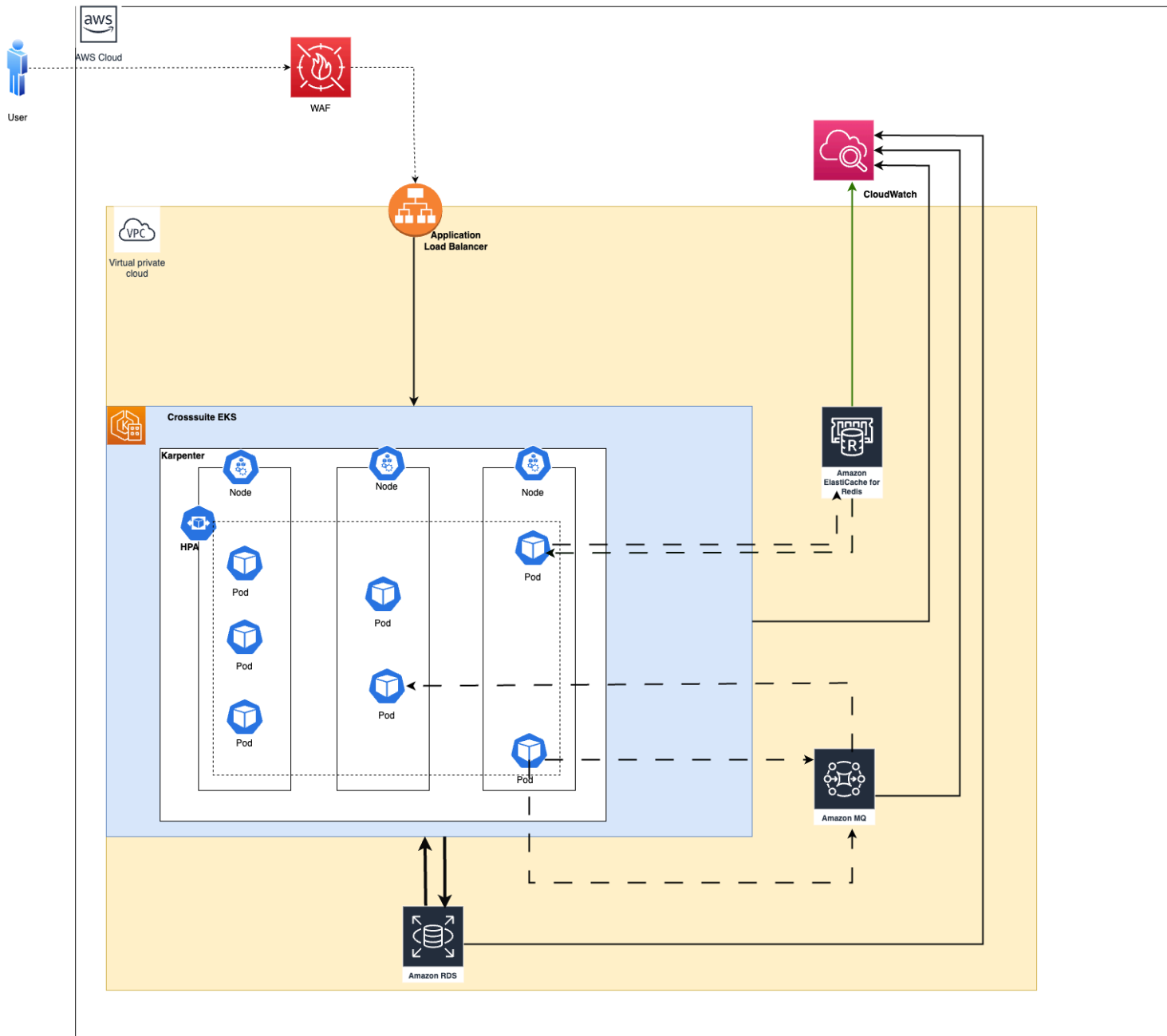
Through this migration and modernization, Crossuite has improved performance and availability. At peak times, its solution can automatically scale to meet demand without affecting the experience of millions of users.



The company has achieved 99.9 percent uptime while reducing latency for its customers. “We can now release code to production anytime without affecting site performance, which was a challenge before,” says Madliak. “On AWS, there’s no service disruption with new releases. That brings us closer to 100 percent availability.”

By migrating to Amazon EKS and Karpenter, Crosssuite also reduced costs. Using Karpenter’s node lifecycle management in combination with cost-optimized instances, Crosssuite reduced its overall costs by 30 percent compared with its previous solution. “Cost reduction has been a huge gain from this migration,” says Pollak. “As a scale-up, we need to focus on cost efficiency.”

Architecture Diagram



Outcome | Increasing User Adoption by 18% and Revenue by 22%

After modernizing its cloud architecture on AWS, Crosssuite has significantly improved the stability, performance, and scalability of its eHealth application. The company now supports a growing number of medical professionals, achieving an 18 percent year-over-year increase in user adoption without compromising application performance. Availability on AWS has greatly increased compared with the on-premises setup. With 22 percent year-over-year

revenue growth, the solution continues to expand its reach and impact. With ongoing enhancements to its features and performance, Crossuite remains at the forefront of medical-practice-management solutions.

“There’s always pressure to move quickly, especially as a scale-up, but with support from the AWS team, we had enough time to modernize right. Planning and support were key to our migration” says Pollak. As Crossuite continues to innovate and expand, it remains committed to leveraging AWS services to maintain its competitive edge and deliver exceptional value to its growing customer base.

About Crossuite

Founded in 2007, Crossuite is a scale-up actively transforming healthcare with fast and intuitive online tools for both patients and healthcare professionals so that they can cooperate to increase the quality of the provided medical services.

Crossuite Team



AWS Services Used

Amazon EKS



Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run Kubernetes in the AWS cloud and on-premises data centers.

[Learn more »](#)

Amazon CloudWatch

Amazon CloudWatch monitors your Amazon Web Services (AWS) resources and the applications you run on AWS in real time. You can use CloudWatch to collect and track metrics, which are variables you can measure for your resources and applications.

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Amazon RDS

Amazon Relational Database Service (Amazon RDS) is an easy-to-manage relational database service optimized for total cost of ownership. It is simple to set up, operate, and scale with demand.

[Learn more »](#)

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